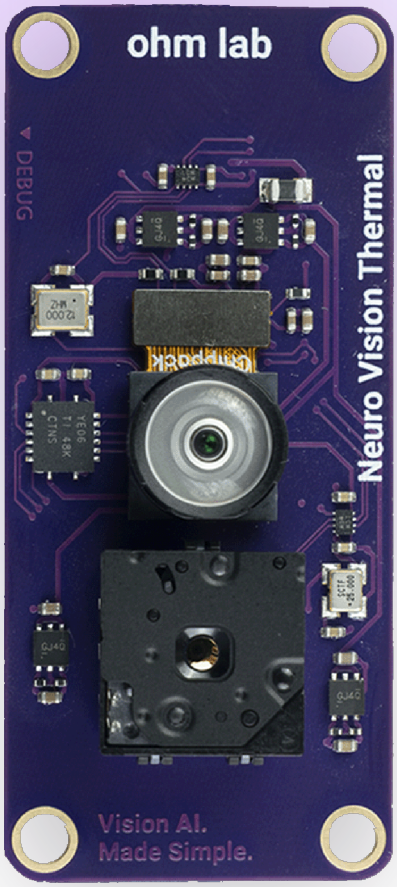
Neuro Vision Thermal

Datasheet



Dual-sensor expansion board for the Neuro N6 that pairs a FLIR Lepton 3.1R radiometric thermal imager with a visible camera for sensor-fusion edge AI applications. The camera connector accepts ST camera promodules; the VD66GY 1.5 MP colour global shutter promodule with a 73° lens is supplied as standard.

160×120
Lepton 3.1R thermal

95°
Thermal field of view

1.5 MP
Standard VD66GY promodule

<50 mK
Thermal sensitivity

PART NUMBER	HARDWARE REVISION	DOCUMENT	DATE
OL-DEV-N6-005	Rev C	OL-DEV-N6-005 Rev D	25 October 2026

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1 Overview

The Neuro Vision Thermal is a dual-sensor expansion board for the Neuro N6. A radiometric FLIR Lepton 3.1R thermal imager and a visible camera on an ST camera promodule view the same scene, so the Neuro N6 can run sensor-fusion applications such as presence detection, thermal anomaly detection and radiometric measurement with visual context.

- FLIR Lepton 3.1R radiometric thermal imager: 160 x 120 pixels, 95° horizontal field of view, longwave infrared 8 to 14 µm.
- Thermal sensitivity below 50 mK, 8.6 Hz effective frame rate, integrated shutter for flat-field correction.
- Radiometric output: an absolute temperature for every pixel, in 0.01 K steps.
- Visible camera on an ST camera promodule connector. Supplied with the VD66GY 1.5 MP colour global shutter promodule: 73° DFOV, up to 1120 x 1364 at 45 fps.
- 2-lane MIPI CSI-2 (visible camera) and SPI VoSPI (thermal) video interfaces.
- Socketed thermal module and pluggable camera promodule, individually replaceable.
- Powered from the Neuro N6 3.3 V and 1.8 V rails through four on-board LDO regulators.
- One 30-pin and one 40-pin board-to-board connector, 3.5 mm mating height.
- Supported by the Neuro N6 Arduino core.

2 Introduction

The Neuro Vision Thermal mounts on the Neuro N6 through two board-to-board connectors with a mating height of 3.5 mm. The thermal imager is radiometric: every pixel reports a factory-calibrated absolute temperature that is independent of the camera temperature, so measurements need no external reference. The visible camera uses a global shutter and captures the whole frame at one instant.

Both sensors are modular. The Lepton 3.1R sits in a Molex socket and the camera promodule plugs into a 30-pin, 0.35 mm pitch connector, so either can be replaced independently of the main board. The camera connector accepts ST camera promodules; the board is supplied with the CAM-6GY-073VIS promodule (VD66GY 1.5 MP colour global shutter sensor, 73° DFOV lens).

The Neuro N6 Arduino core, available from neuron6.com, provides drivers and examples for both cameras.

For detailed information on the Lepton 3.1R and the ST camera promodules, refer to the Teledyne FLIR and STMicroelectronics documentation.

2.1 Ordering information

PRODUCT	PART NUMBER	HARDWARE REVISION
Neuro Vision Thermal	`OL-DEV-N6-005`	Rev C

Table 1. Ordering information. Quote the part number and hardware revision when ordering and when contacting Ohm Lab for support.

2.2 Related products

- Neuro N6 (OL-DEV-N6-001)
- Neuro Vision OV5640 (OL-DEV-N6-002)
- Neuro Vision OV5640-W (OL-DEV-N6-003)
- Neuro Vision ST Cam (OL-DEV-N6-004)
- Neuro Vision TFT (OL-DEV-N6-006)
- Neuro Vision TFT-W (OL-DEV-N6-007)
- Neuro N6 Enclosure (OL-DEV-N6-009)
- Neuro N6 Debug Adapter (OL-DEV-N6-010)
- B-CAMS-IMX Adapter (OL-DEV-N6-011)

3 Features

FEATURE	DESCRIPTION
Thermal imaging	
FLIR Lepton 3.1R	160 x 120 pixel uncooled VOx microbolometer (FLIR part 500-0758-03); radiometric per-pixel temperature output in 0.01 K steps; 95° horizontal, 119° diagonal field of view; 1.23 mm focal length; 8 to 14 µm; thermal sensitivity below 50 mK; 8.6 Hz effective frame rate; integrated shutter for flat-field correction, which the Arduino core can also trigger; socketed, individually replaceable
Visible camera	
Camera promodule	ST camera promodule on a 30-pin, 0.35 mm pitch connector. Standard CAM-6GY-073VIS: VD66GY 1.5 MP colour global shutter, 73° DFOV. With the Arduino core: 320 x 240 and 640 x 480 at 88 fps, 1024 x 768 and 1120 x 720 at 60 fps, 1120 x 1364 at 45 fps; RAW8 demosaiced by the Neuro N6 image signal processor; automatic or manual exposure, gain 1 to 64x, fixed, automatic or manual white balance; 2-lane MIPI CSI-2 at 804 Mbit/s per lane; I2C control at 1.8 V
Interfaces	
Board-to-board connectors	40-pin and 30-pin HC-PBB40C receptacles, 0.4 mm pitch, 3.5 mm mating height: 2-lane MIPI CSI-2, SPI VoSPI thermal video, I2C (3.3 V host side), thermal module reset and power-down, NRST
Sensor connectors	Molex 105028-1001 socket for the Lepton module; 30-pin 0.35 mm board-to-board connector for the camera promodule
Power	
Supply	3.3 V and 1.8 V from the Neuro N6; four on-board LDO regulators (2.8 V camera analog and Lepton I/O, 1.15 V camera core, 2.8 V Lepton analog, 1.2 V Lepton core); ferrite-filtered supply input

Table 2. Neuro Vision Thermal features.

4 Mechanical

The board follows the Adafruit Feather outline with the four mounting holes on the 45.72 x 17.78 mm pattern. All dimensions are in millimetres and are taken from the PCB design data and the 3D model. The board mounts back to back with the Neuro N6, so the camera side faces away from the Neuro N6.

PCB outline	50.80 x 22.86 mm, corner radius 2.54 mm
PCB	1.6 mm
Height above the PCB	7.2 mm (Lepton 3.1R seated in its Molex 105028-1001 socket, Teledyne FLIR figure; camera promodule 3.3 mm)
Height below the PCB	3.4 mm (board-to-board receptacles)
Overall height	12.2 mm
Stack height on the Neuro N6	19.4 mm from the Neuro N6 battery connector to the top of the expansion board, with a 3.5 mm mating gap
Mounting holes	4 x Ø2.54 mm plated, 45.72 x 17.78 mm pattern, 2.54 mm from the board edges, connected to GND
Board-to-board receptacles	40-pin centred 17.61 mm and 30-pin centred 12.37 mm from the Feather body edge of the Neuro N6 when mated; both on the board centre line

Table 3. Mechanical summary.

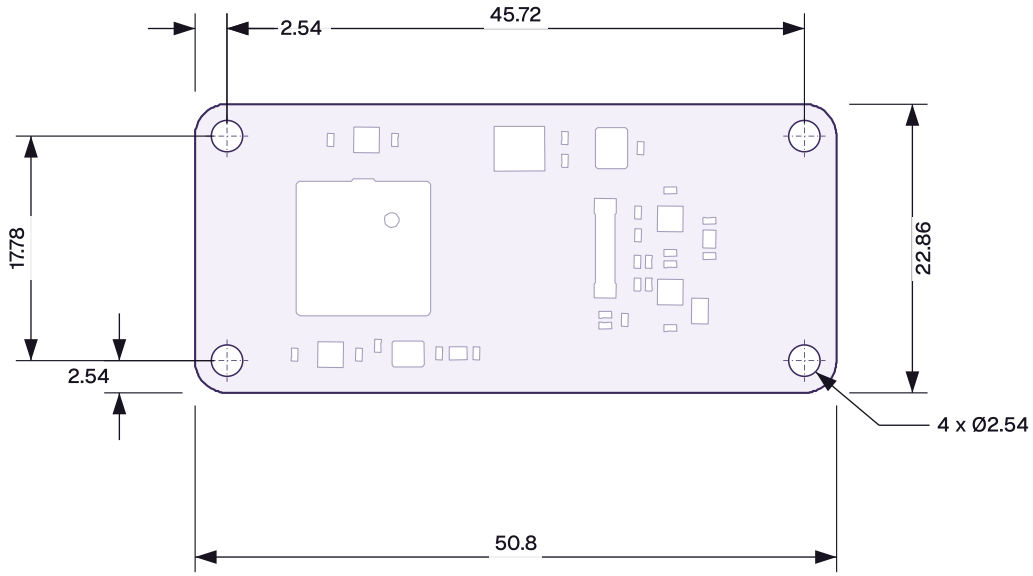


Figure 1. Neuro Vision Thermal top view (component side). Component outlines are taken from the 3D model; dimensions are nominal design values in millimetres.

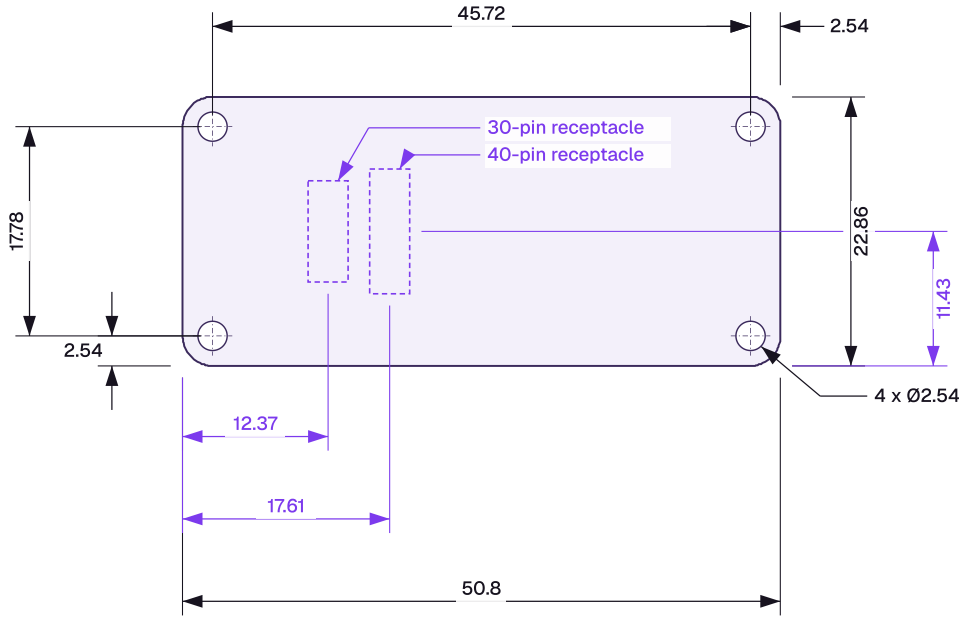


Figure 2. Neuro Vision Thermal bottom view (mirrored, as seen from below), showing the board-to-board receptacles. Positions match the plug positions on the Neuro N6.

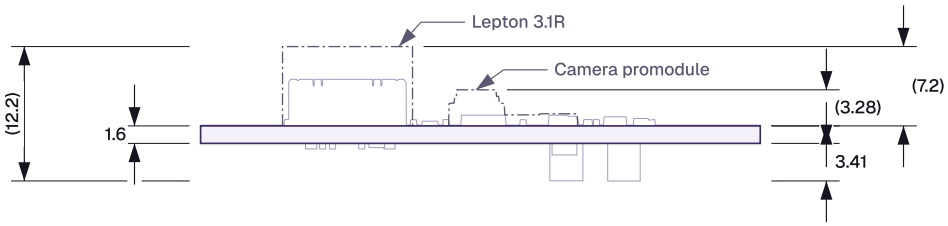


Figure 3. Neuro Vision Thermal side view. The Lepton 3.1R and the camera promodule are not part of the 3D model and are shown as phantom outlines; dimensions in brackets are reference values from the Teledyne FLIR and STMicroelectronics data.

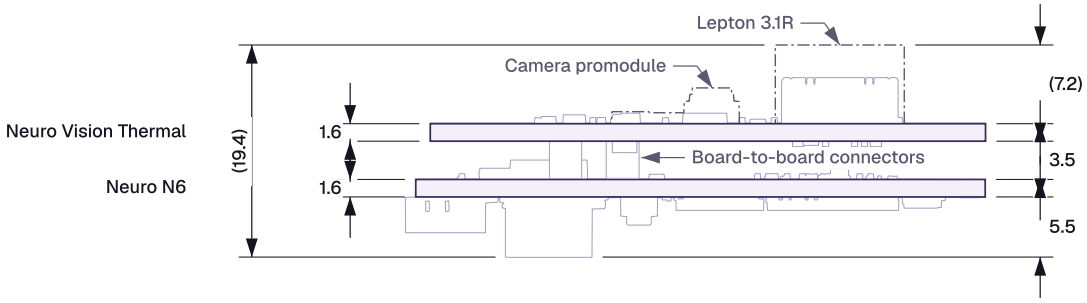


Figure 4. Neuro Vision Thermal mounted on the Neuro N6, side view to scale from the two 3D models. The boards mount back to back with a 3.5 mm mating gap; the Neuro N6 component side faces down. The Lepton 3.1R and the camera promodule are shown as phantom outlines.

5 Block diagram

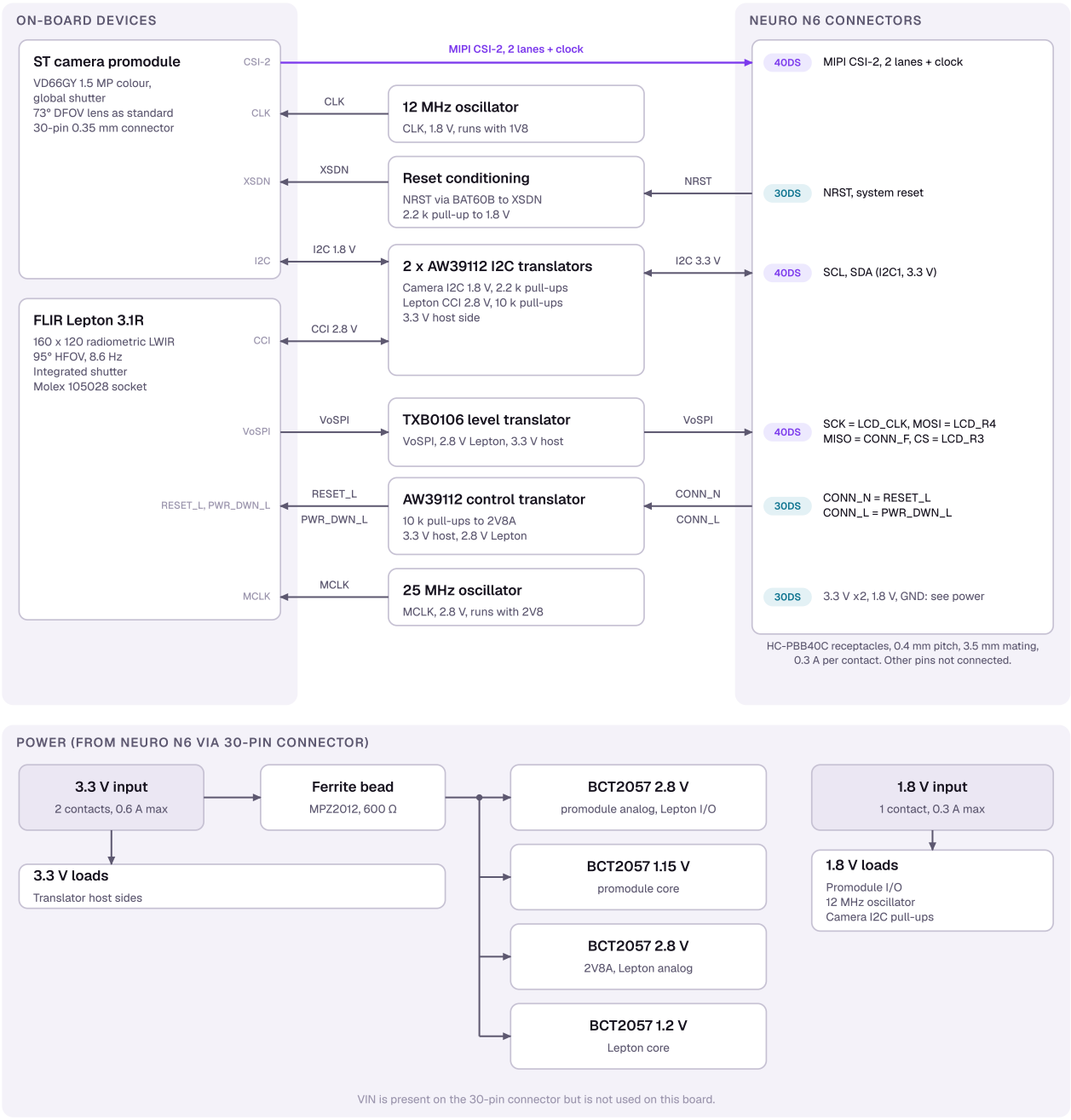


Figure 5. Neuro Vision Thermal block diagram.

6 Hardware interfaces

6.1 Thermal imaging module

The FLIR Lepton 3.1R sits in a Molex 105028 socket. It images longwave infrared radiation (8 to 14 µm) on a 160 x 120 uncooled microbolometer array and, being radiometric, outputs a factory-calibrated absolute temperature for every pixel. The integrated shutter periodically performs a flat-field correction, during which a quiet click may be heard and the image is briefly frozen; the Arduino core can also trigger one.

Thermal video is streamed over the Lepton VoSPI protocol on a 4-wire SPI bus, level-translated by a TXB0106 between the 3.3 V host domain and the 2.8 V module domain. Control uses the Lepton camera command interface (CCI), an I2C-compatible bus at 2.8 V. The 25 MHz master clock comes from an on-board oscillator, and the module RESET_L and PWR_DWN_L inputs are driven by Neuro N6 GPIO through an AW39112 level translator.

> **Caution** The thermal module contains precision optics. Handle the board by its edges, press only on the module housing when seating or removing it, and do not touch the lens.

6.2 Camera module

The visible camera is an ST camera promodule on a 30-pin, 0.35 mm pitch board-to-board connector. The board is supplied with the CAM-6GY-073VIS, which carries the VD66GY colour global shutter sensor and a 73° DFOV lens. Global shutter operation exposes all pixels at the same instant, which suits fast-moving scenes and capture synchronised with the thermal imager. The sensor outputs RAW8 frames on the 2-lane MIPI CSI-2 interface routed to the 40-pin connector, which the Neuro N6 image signal processor demosaics, and is clocked by an on-board 12 MHz oscillator. The promodule STROBE and GPIO signals are not connected on this board.

6.3 Connectors

CONNECTOR	PART NUMBER	MATING PART
Thermal module socket	`105028-1001` (Molex)	FLIR Lepton module
Camera module connector	`BM28B0.6-30DS/2-0.35V(51)`	`BM28B0.6-30DP/2-0.35V(51)` (on the module)
30-pin board-to-board receptacle	`HC-PBB40C-30DS-0.4V-3.5-02`	`HC-PBB40C-30DP-0.4V-02` (Neuro N6)
40-pin board-to-board receptacle	`HC-PBB40C-40DS-0.4V-3.5-02`	`HC-PBB40C-40DP-0.4V-02` (Neuro N6)

Table 4. Connector part numbers.

The board-to-board receptacles mate with the plugs fitted to the underside of the Neuro N6 at a mating height of 3.5 mm. The two boards mount back to back, so the component sides of both boards face outwards. Align both connectors before pressing the boards together, and separate the boards evenly when removing the expansion board.

7 Board operation

7.1 Absolute maximum ratings

Absolute maximum ratings are stress ratings only. They are not operating conditions, and functional operation at or beyond these limits is neither implied nor guaranteed. Exposure to conditions at the limits may reduce reliability. Each limit is the lowest rating among the components connected to that net on the Neuro Vision Thermal schematic. All voltages are referenced to GND.

PARAMETER	MIN	MAX	UNIT	LIMITING ELEMENT
Supply voltage, VDD_3V3	-0.3	6.3	V	6.3 V ceramic capacitors on the 3V3 net
Supply voltage, VDD_1V8	-0.3		V	Camera promodule I/O supply: refer to the ST promodule datasheet
Voltage on host-side 3.3 V signals (SCL, SDA, SPI, control)	-0.3	VDD_3V3 + 0.3	V	TXB0106 and AW39112 B ports
Voltage on NRST	-0.3	VDD_3V3 + 0.3	V	Neuro N6 reset line at 3.3 V; BAT60B on this board
Voltage on camera-domain 1.8 V signals (SCL1V8, SDA1V8)	-0.3	VDD_1V8 + 0.3	V	AW39112 A port, promodule
Continuous current per board-to-board contact		0.3	A	HC-PBB40C 0.4 mm pitch contact
Storage temperature	-40	80	°C	Lepton 3.1R

Table 5. Absolute maximum ratings.

The 3.3 V rail reaches the board on two contacts of the 30-pin connector and the 1.8 V rail on one, giving 0.6 A and 0.3 A respectively at the 0.3 A per-contact limit. The VIN pins on the 30-pin connector are not connected on this board.

7.2 Recommended operating conditions

PARAMETER	MIN	MAX	UNIT	CONDITIONS
Ambient temperature	0	30	°C	Indoor, dry locations only
Relative humidity	25	75	%	Non-condensing
Supply				From the Neuro N6, powered with 5 V DC via USB-C or its battery

Table 6. Recommended operating conditions.

7.3 Electrical characteristics

PARAMETER	MIN	TYP	MAX	UNIT	CONDITIONS
Supply voltage, VDD_3V3	3.23	3.33	3.44	V	From the Neuro N6
Supply voltage, VDD_1V8	1.75	1.8	1.85	V	From the Neuro N6
On-board camera analog rail, 2V8		2.8		V	BCT2057, also Lepton I/O and 25 MHz oscillator
On-board camera core rail, 1V15		1.15		V	BCT2057
On-board thermal analog rail, 2V8A		2.8		V	BCT2057, Lepton VDD and control pull-ups
On-board thermal core rail, 1V2		1.2		V	BCT2057, Lepton VDDC
Camera-domain logic level		1.8		V	Camera I2C and shutdown
Thermal-domain logic level		2.8		V	VoSPI, CCI, reset, power-down
Host-side logic level		3.3		V	I2C, SPI, control
Lepton power consumption		150	800	mW	Typical operating; up to 800 mW during a shutter event

Table 7. Electrical characteristics.

7.4 Power supply

The board has no power input of its own. It is powered from the Neuro N6 through the board-to-board connectors, which provide the 3.3 V (VDD_3V3) and 1.8 V (VDD_1V8) rails. The 3.3 V rail is filtered by a ferrite bead and feeds four BCT2057 low-dropout regulators: 2.8 V and 1.15 V supplies for the camera analog and core domains, and independent 2.8 V and 1.2 V supplies for the thermal module. The camera I/O domain, the 12 MHz oscillator and the 1.8 V I2C pull-ups run from the 1.8 V rail; the thermal module I/O, the 25 MHz oscillator and the sensor side of the level translators run from the on-board 2.8 V rail.

RAIL	SOURCE	VOLTAGE	SUPPLIES
`VDD_3V3`	Neuro N6, 30-pin connector	3.3 V	Host side of the level translators, LDO inputs (via ferrite bead)
`VDD_1V8`	Neuro N6, 30-pin connector	1.8 V	Camera I/O, 12 MHz oscillator, 1.8 V I2C pull-ups
`2V8`	On-board LDO	2.8 V	Camera analog supply, Lepton I/O, 25 MHz oscillator, sensor side of the translators
`2V8A`	On-board LDO	2.8 V	Lepton analog supply and control-signal pull-ups
`1V15`	On-board LDO	1.15 V	Camera digital core
`1V2`	On-board LDO	1.2 V	Lepton digital core
`VIN`	Neuro N6, 30-pin connector	VBUS or VBAT	Not used on this board

Table 8. Power rails.

> **Warning** Do not attach or detach the Neuro Vision Thermal while the Neuro N6 is powered. Disconnect the USB-C supply and the battery before changing expansion boards; otherwise the boards may be permanently damaged.

7.5 Reset and clocking

The system reset signal NRST from the Neuro N6 controls the camera: the sensor's active-low XSDN (shutdown) input is pulled up to the 1.8 V rail through 2.2 kΩ and is pulled low through a BAT60B diode whenever NRST is asserted. The thermal module is not connected to NRST; its active-low RESET_L and PWR_DWN_L inputs are pulled up to 2V8A through 10 kΩ and are driven by Neuro N6 GPIO through a level translator, so the host resets or powers down the imager under software control. Two on-board oscillators provide the sensor clocks: 12 MHz for the camera (from the 1.8 V rail) and 25 MHz for the thermal module (from the 2.8 V rail). Both run whenever their supply is present.

8 Peripheral interfaces

Connector pins are given as `40DS:n` and `30DS:n`, the receptacle pin numbers on this board. The matching Neuro N6 plug pin is listed in the connector tables. The Neuro N6 datasheet gives the MCU pin for every signal.

8.1 MIPI CSI-2

The promodule 2-lane MIPI CSI-2 interface is routed differentially from the promodule connector to the 40-pin connector, matching the Neuro N6 D-PHY receiver (CSI-2 version 1.3 compatible). The Arduino core runs the VD66GY at 804 Mbit/s per lane. Ground guard pins separate the lanes. On this board the camera data lanes are swapped at the promodule connector; the Arduino core sets the Neuro N6 receiver lane mapping to match.

NET	CONNECTOR PIN	DESCRIPTION
`CSI_D0P`	`40DS:31`	Data lane 0 positive
`CSI_D0N`	`40DS:29`	Data lane 0 negative
`CSI_CLKP`	`40DS:23`	Clock lane positive
`CSI_CLKN`	`40DS:21`	Clock lane negative
`CSI_D1P`	`40DS:15`	Data lane 1 positive
`CSI_D1N`	`40DS:13`	Data lane 1 negative

Table 9. MIPI CSI-2 pins.

8.2 I2C

The Neuro N6 I2C1 bus enters the board at 3.3 V on the 40-pin connector, from the bus that also serves the on-board IMU and magnetometer of the Neuro N6. Two AW39112 translators split it into a 1.8 V bus for the camera (2.2 kΩ pull-ups) and a 2.8 V bus for the Lepton camera command interface (10 kΩ pull-ups). Standard mode (100 kHz) and fast mode (400 kHz) are supported.

NET	LOGIC LEVEL	CONNECTOR PIN	DESCRIPTION
`SCL`	3.3 V	`40DS:35`	I2C clock, host side
`SDA`	3.3 V	`40DS:37`	I2C data, host side
`SCL1V8`	1.8 V		I2C clock, camera domain
`SDA1V8`	1.8 V		I2C data, camera domain
`SCL_2V8`	2.8 V		CCI clock, thermal domain
`SDA_2V8`	2.8 V		CCI data, thermal domain

Table 10. I2C pins.

DEVICE	7-BIT ADDRESS	NOTES
Promodule image sensor	`0x10`	VD66GY on the standard promodule; refer to STMicroelectronics for other promodules
Lepton 3.1R (CCI)	`0x2A`	Camera command interface

Table 11. I2C addresses.

8.3 SPI (thermal video)

Thermal video is streamed over the Lepton VoSPI protocol on a 4-wire SPI bus (SPI mode 3, clock up to 20 MHz). The bus is level-translated between the 3.3 V host domain and the 2.8 V thermal domain by a TXB0106. Three LCD interface pins and CONN_F on the 40-pin connector carry the signals; the Neuro N6 reads them on SPI1 (SCK PA5, MISO PB4), with chip select on PF4.

NET	LEPTON SIGNAL	CONNECTOR PIN	DESCRIPTION
LCD_R3	CS_N	40DS:2	SPI chip select, active low
LCD_CLK	SCK	40DS:14	SPI clock
CONN_F	MISO	40DS:26	SPI data, module to host
LCD_R4	MOSI	40DS:36	SPI data, host to module

Table 12. SPI pins.

8.4 Camera control signals

NET	PROMODULE SIGNAL	CONNECTOR PIN	DESCRIPTION
NRST	XSDN	30DS:29	Active-low camera shutdown, driven by the system NRST (2.2 kΩ pull-up to 1.8 V)
CLK	CLK		12 MHz master clock from the on-board oscillator
STROBE , GPIO1..5			Not connected on this board

Table 13. Camera control signals.

8.5 Thermal control signals

NET	LEPTON SIGNAL	CONNECTOR PIN	DESCRIPTION
CONN_N	RESET_L	30DS:6	Active-low module reset, 10 kΩ pull-up, level translated
CONN_L	PWR_DWN_L	30DS:8	Active-low module power-down, 10 kΩ pull-up, level translated
LEPTONCLK	MASTER_CLK		25 MHz master clock from the on-board oscillator

Table 14. Thermal control signals.

8.6 40-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	SD_D1	40	SDMMC (not used on this board)
2	LCD_R3	39	Thermal SPI chip select (3.3 V)
3	SD_CMD	38	SDMMC (not used on this board)
4	LCD_R7	37	LCD interface (not used on this board)
5	SD_CK	36	SDMMC (not used on this board)
6	LCD_R6	35	LCD interface (not used on this board)
7	SD_D3	34	SDMMC (not used on this board)
8	SD_D0	33	SDMMC (not used on this board)

PIN	NET	N6 PLUG PIN	FUNCTION
9	GND	32	Ground
10	LCD_VSYNC	31	LCD interface (not used on this board)
11	GND	30	Ground
12	CONN_G	29	GPIO (not used on this board)
13	CSI_D1N	28	MIPI CSI-2 data lane 1 negative
14	LCD_CLK	27	Thermal SPI clock (3.3 V)
15	CSI_D1P	26	MIPI CSI-2 data lane 1 positive
16	LCD_B5	25	LCD interface (not used on this board)
17	GND	24	Ground
18	LCD_B3	23	LCD interface (not used on this board)
19	GND	22	Ground
20	LCD_B4	21	LCD interface (not used on this board)
21	CSI_CLKN	20	MIPI CSI-2 clock lane negative
22	LCD_B6	19	LCD interface (not used on this board)
23	CSI_CLKP	18	MIPI CSI-2 clock lane positive
24	LCD_B7	17	LCD interface (not used on this board)
25	GND	16	Ground
26	CONN_F	15	Thermal SPI MISO (3.3 V)
27	GND	14	Ground
28	LCD_G5	13	LCD interface (not used on this board)
29	CSI_D0N	12	MIPI CSI-2 data lane 0 negative
30	LCD_G3	11	LCD interface (not used on this board)
31	CSI_D0P	10	MIPI CSI-2 data lane 0 positive
32	LCD_G7	9	LCD interface (not used on this board)
33	GND	8	Ground
34	LCD_G6	7	LCD interface (not used on this board)
35	SCL	6	I2C clock, 3.3 V host side (Neuro N6 I2C1)
36	LCD_R4	5	Thermal SPI MOSI (3.3 V)
37	SDA	4	I2C data, 3.3 V host side (Neuro N6 I2C1)
38	CONN_C	3	GPIO (not used on this board)
39	SD_D2	2	SDMMC (not used on this board)
40	CONN_D	1	GPIO (not used on this board)

Table 15. 40-pin receptacle pin assignment.

8.7 30-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	LCD_HSYNC	29	LCD interface (not used on this board)
2	SAI1_MCLK_B	30	Audio interface (not used on this board)
3	CONN_J	27	GPIO (not used on this board)
4	LCD_R5	28	LCD interface (not used on this board)
5	CONN_H	25	GPIO (not used on this board)
6	CONN_N	26	Thermal module reset (RESET_L), active low
7	CONN_M	23	GPIO (not used on this board)
8	CONN_L	24	Thermal module power-down (PWR_DWN_L), active low
9	CONN_E	21	GPIO (not used on this board)
10	DEBUG_CLK	22	SWD clock (not used on this board)
11	LCD_DE	19	LCD interface (not used on this board)
12	DEBUG_DAT	20	SWD data (not used on this board)
13	VIN	17	VIN rail from the Neuro N6 (not used on this board)
14	LCD_G4	18	LCD interface (not used on this board)
15	VIN	15	VIN rail from the Neuro N6 (not used on this board)
16	SAI1_SCK_B	16	Audio interface (not used on this board)
17	VIN	13	VIN rail from the Neuro N6 (not used on this board)
18	SAI1_SD_B	14	Audio interface (not used on this board)
19	VIN	11	VIN rail from the Neuro N6 (not used on this board)
20	CONN_K	12	GPIO (not used on this board)
21	VIN	9	VIN rail from the Neuro N6 (not used on this board)
22	CONN_B	10	GPIO (not used on this board)
23	3V3	7	3.3 V supply input
24	GND	8	Ground
25	3V3	5	3.3 V supply input
26	LCD_G2	6	LCD interface (not used on this board)
27	1V8	3	1.8 V supply input
28	SAI1_FS_B	4	Audio interface (not used on this board)
29	NRST	1	System reset input (shuts down the camera)
30	CONN_A	2	GPIO (not used on this board)

Table 16. 30-pin receptacle pin assignment.

9 Regulatory compliance

This section describes the regulatory status of the Neuro Vision Thermal as supplied by Ohm Lab. The board is an expansion board intended to be operated only when mounted on a Neuro N6, and it is assessed as part of that combination rather than as a stand-alone product. Where the board is incorporated into a finished product, responsibility for the compliance of that product rests with its manufacturer.

9.1 Product marking

SYMBOL	MARKING	APPLIED TO	MEANING
CE	CE	Board and packaging	Conformity with the EMC Directive 2014/30/EU and the RoHS Directive 2011/65/EU.
	Crossed-out wheeled bin	Board and packaging	Do not dispose of with unsorted household waste. The bar identifies equipment placed on the market after 13 August 2005.
	Triman and Info-tri	Packaging	France: sorting information required by the French environment code. The Triman is printed on the carton; an Info-tri sticker with the sorting instructions is added to orders for France.

Table 17. Product markings. The controlled artwork is held in the technical file.

No UKCA marking is applied: CE marking is accepted for goods placed on the Great Britain market. RoHS conformity is declared through the CE marking and carries no separate mark.

9.2 Electromagnetic compatibility

The Neuro N6 platform, including the Neuro Vision Thermal, has been assessed for electromagnetic compatibility as a product family, based on tests of worst-case configurations of the Neuro N6 with expansion boards fitted. The Neuro Vision Thermal conforms to the EMC Directive 2014/30/EU, with emissions assessed to EN 55032 Class B and immunity to EN 55035. The assessment applies to the board mounted on a Neuro N6 and operated with the default clock settings of the Neuro N6 Arduino core. The declaration of conformity is available from Ohm Lab on request.

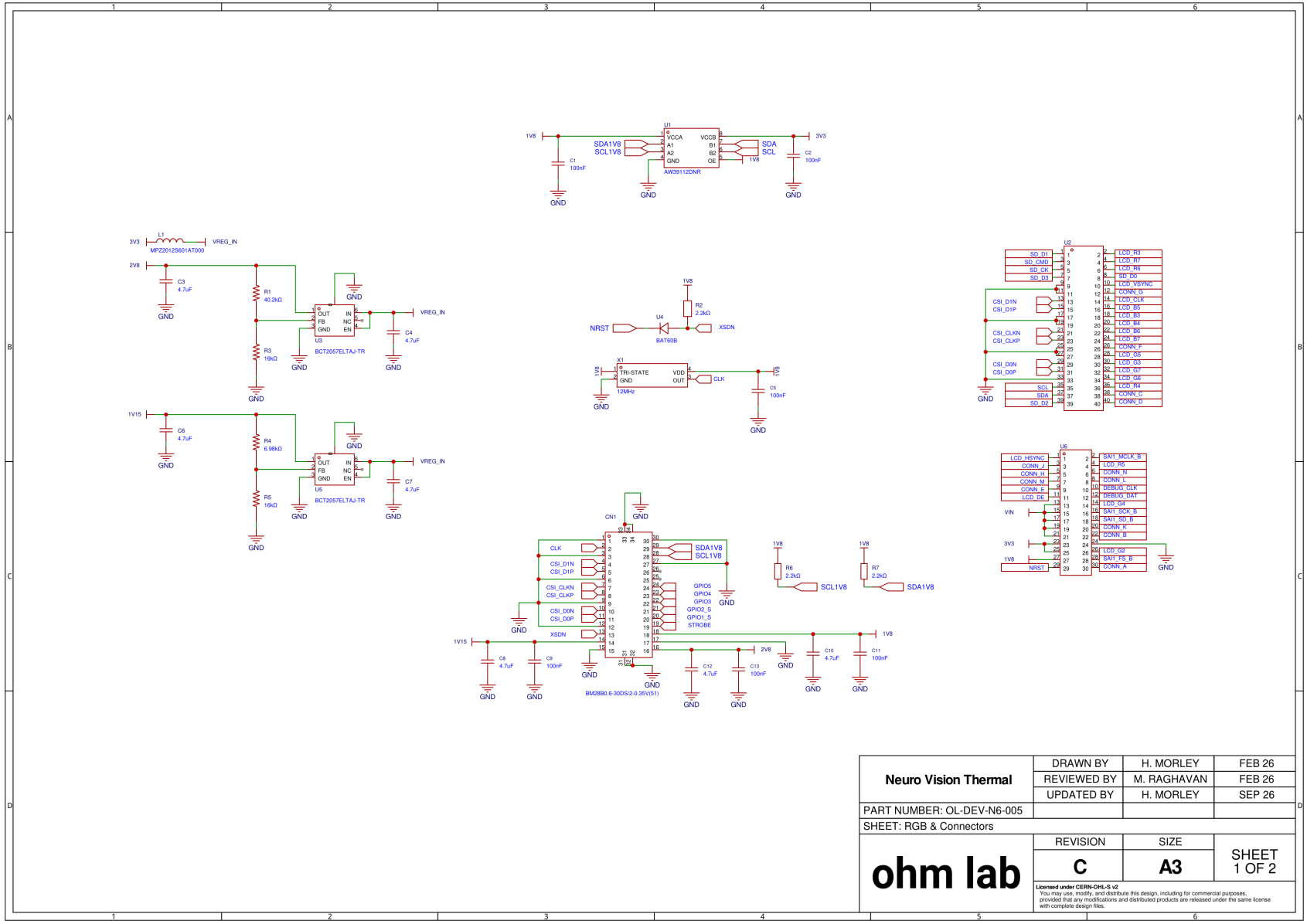
9.3 Materials and disposal

The Neuro Vision Thermal is manufactured to the substance restrictions of Directive 2011/65/EU (RoHS 2) as amended and the UK Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012. Supplier material declarations, and declarations covering substances of very high concern under REACH, are held in the technical file.

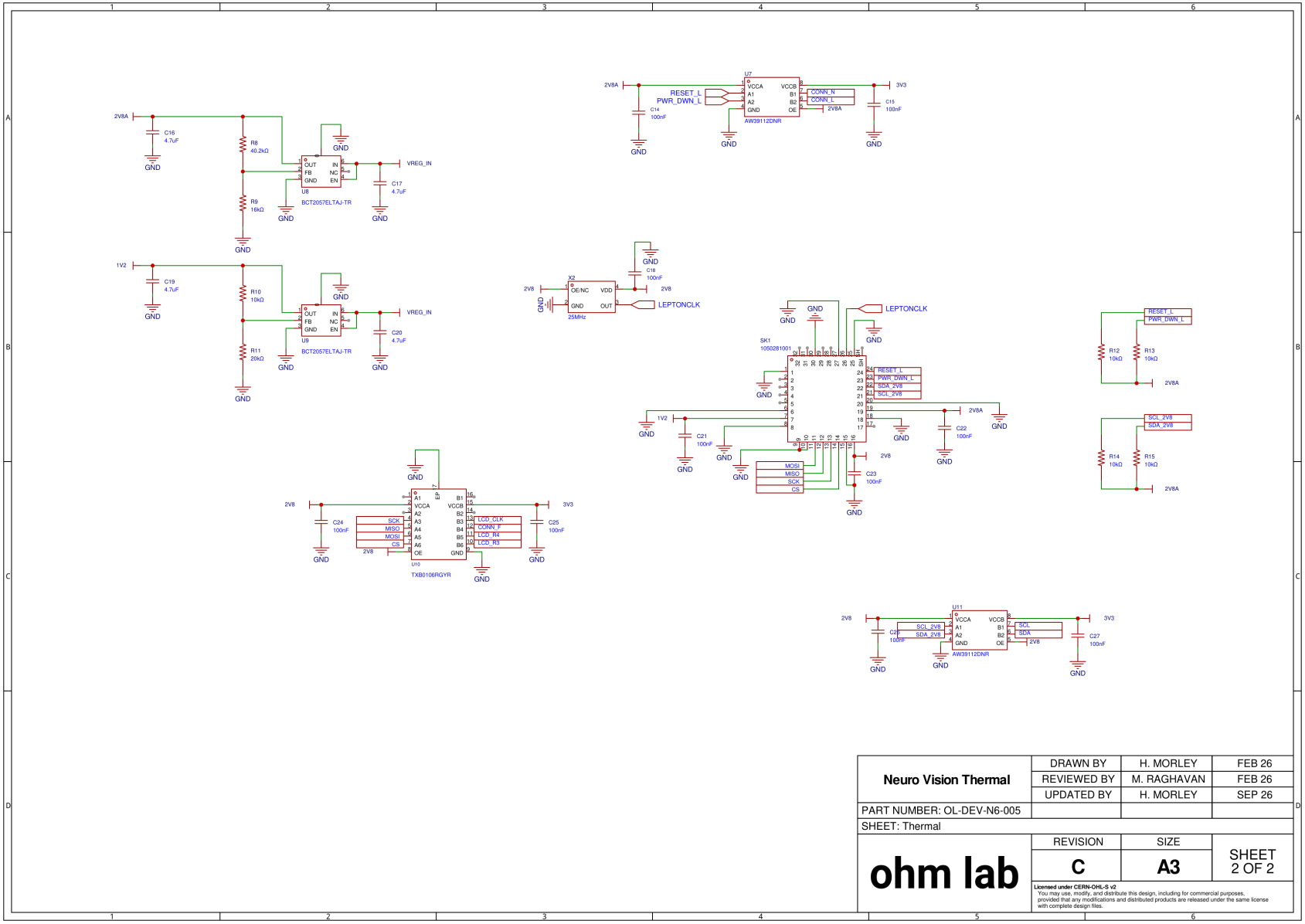
At end of life, do not dispose of the board with unsorted household waste. Return it to a designated collection point for electrical and electronic equipment, or to Ohm Lab.

10 Schematics

The schematics are reproduced on the following pages for reference. Net names match those used in this document. Full-resolution sheets are published in the Neuro N6 Hardware repository (github.com/ohmlab-ltd/Neuro-N6-Hardware).



Neuro Vision Thermal Rev C schematic, sheet 1 of 2: visible camera, power and connectors. XSDN pull-up R2 shown at 2.2 kΩ, as fitted.



Neuro Vision Thermal Rev C schematic, sheet 2 of 2: thermal module, level translators and control.

11 Document revision history

REVISION	DATE	CHANGES
A	02/07/2026	• Pre-production first release
B	02/09/2026	• Ratings and regulatory section added
C	24/09/2026	• Rev C hardware • New block diagram and drawings
D	25/10/2026	• Features as supported by the Arduino core • XSDN pull-up 2.2 kΩ

Table 18. Document revision history.